

**AP Calculus II**  
**Notes 9.1**  
**Sequences**

A sequence is any function whose domain is the set of positive integers. However, it is common to represent sequences in subscript notation, as shown below:

2, 4, 6, 8, ..., n, ...

the entire sequence is denoted as:

$a_1, a_2, a_3, a_4, \dots, a_n, \dots$   
 first term      second term      ...       $n^{\text{th}}$  term

$\{a_n\}$

Writing out the Terms of a Sequence

**Ex. 1:** Write out the first five terms of the following sequences.

a)  $\{a_n\} = \{3 + (-1)^n\}$       Start at  $n=1$

b)  $\{b_n\} = \left\{\frac{n}{1+3n}\right\}$

$a_1 = 3 + (-1)^1 = 2$

$a_2 = 3 + (-1)^2 = 4$

$a_3 = 3 + (-1)^3 = 2$

$a_4 = 3 + (-1)^4 = 4$

$a_5 = 3 + (-1)^5 = 2$

$\boxed{2, 4, 2, 4, 2, \dots}$

$\boxed{\frac{1}{4}, \frac{2}{7}, \frac{3}{10}, \frac{4}{13}, \frac{5}{16}, \dots}$

c)  $\{e_n\} = \left\{\frac{(-1)^{n+1}}{n!}\right\}$

d)  $\{c_n\} = \left\{\frac{n^2}{2^{n-1}}\right\}$

factorial

$7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 5040$

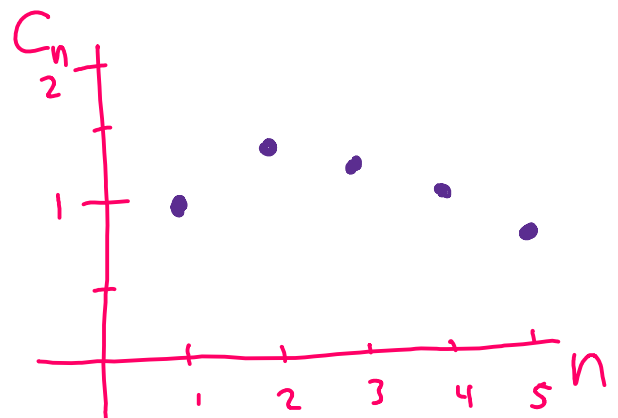
$6! = 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 720$

$\frac{(-1)^2}{1!}, \frac{(-1)^3}{2!}, \frac{(-1)^4}{3!}, \frac{(-1)^5}{4!}, \frac{(-1)^6}{5!}$

$\boxed{1, -\frac{1}{2}, \frac{1}{6}, -\frac{1}{24}, \frac{1}{120}, \dots}$

\*  $0! = 1$  \* recognize the  $n!$  pattern

$\boxed{1, \frac{4}{3}, \frac{9}{7}, \frac{16}{15}, \frac{25}{31}, \dots}$



## Limit of a Sequence

One major purpose of sequences is to see if the terms approach a certain value as the terms go on forever. These sequences are said to **converge**. For example:

The sequence  $\left\{\frac{1}{2^n}\right\} = \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32}, \dots$  converges to 0  
each terms gets closer & closer to 0

## Definition of the Limit of a Sequence

Let  $L$  be a real number. The limit of a sequence  $\{a_n\}$  is  $L$ , written as  $\lim_{n \rightarrow \infty} a_n = L$ , and if this limit exists, then the sequence **converges** to  $L$ . If the limit of a sequence does not exist, then the sequence **diverges**.

Additionally, let  $L$  be a real number. Let  $f$  be a function of a real variable such that  $\lim_{x \rightarrow \infty} f(x) = L$ . If  $\{a_n\}$  is a sequence such that  $f(n) = a_n$  for every positive integer  $n$ , then  $\lim_{n \rightarrow \infty} a_n = L$ .

## Determining Convergence or Divergence

**Ex. 2:** Determine whether each of the following sequences converge or diverge.

a)  $\{a_n\} = \{3 + (-1)^n\}$

$\lim_{n \rightarrow \infty} (3 + (-1)^n) \rightarrow (-1)^\infty$   
will keep oscillating from  
-1 to 1 to -1 to 1. ....

so the terms go from  
2, 4, 2, 4, 2, 4, 2, ...

Doesn't approach a  
single number  $L$ , so

Sequence Diverges

b)  $\{b_n\} = \left\{\frac{n}{1+3n}\right\}$

$\lim_{n \rightarrow \infty} \frac{n}{1+3n} \rightarrow \frac{\infty}{\infty}$  L'Hôp!

$\lim_{n \rightarrow \infty} n \rightarrow \infty, \lim_{n \rightarrow \infty} 1+3n \rightarrow \infty$   
 $\therefore$  L'Hôp applies

$\therefore \lim_{n \rightarrow \infty} \frac{1}{3} = \boxed{\frac{1}{3}}$

Sequence converges  
to  $\frac{1}{3}$

$\cos \pi = -1, \cos 2\pi = 1, \cos 3\pi = -1 \dots$  just  $(-1)^n$ !!!

c)  $\left\{ \frac{3^n + \cos}{2n^2} \right\}$

$$\lim_{n \rightarrow \infty} \frac{3^n + (-1)^n}{2n^2}$$

As  $n \rightarrow \infty$ , the alternating  $(-1)^n$  is very insignificant. Imagine  $3^{100} \pm 1 \dots$  so ignore it!!

$$\lim_{n \rightarrow \infty} 3^n \rightarrow \infty, \lim_{n \rightarrow \infty} 2n^2 \rightarrow \infty \text{ L'Hôp!}$$

$$= \lim_{n \rightarrow \infty} \frac{3^n \ln 3}{4n} \rightarrow \infty \text{ L'Hôp again!!}$$

$$= \lim_{n \rightarrow \infty} \frac{3^n (\ln 3)^2}{4} \rightarrow \frac{\infty}{4} \quad \boxed{\text{Diverges}}$$

e)  $\left\{ \cos\left(\frac{3}{n}\right) \right\}$

$$\lim_{n \rightarrow \infty} \cos\left(\frac{3}{n}\right) \rightarrow$$

$$\cos\left(\frac{3}{\infty}\right) \rightarrow \cos 0$$

$$= \boxed{1}$$

Converges!

d)  $\left\{ \frac{n^2}{e^{n-1} + 1} \right\}$

$$\lim_{n \rightarrow \infty} \frac{n^2}{e^{n-1} + 1}$$

$$\lim_{n \rightarrow \infty} n^2 = \infty, \lim_{n \rightarrow \infty} e^{n-1} + 1 = \infty$$

$\therefore$  L'Hôp applies

$$= \lim_{n \rightarrow \infty} \frac{2n}{e^{n-1}}$$

$$\lim_{n \rightarrow \infty} 2n = \infty, \lim_{n \rightarrow \infty} e^{n-1} = \infty$$

$\therefore$  L'Hôp again!

$$= \lim_{n \rightarrow \infty} \frac{2}{e^{n-1}} \Rightarrow \frac{2}{\infty} \rightarrow 0$$

$\boxed{\text{converges}}$

f)  $\left\{ \frac{(-1)^n n!}{(n+2)!} \right\}$

We don't know derivative of  $n!$

so L'Hôp is useless  $\therefore$

Let's use simplification though!

$$\frac{(-1)^n n!}{(n+2)!} = \frac{(-1)^n \overbrace{(n)(n-1)(n-2) \dots (2)(1)}^{n!}}{\underbrace{(n+2)(n+1)(n)(n-1) \dots (2)(1)}_{(n+2)!}}$$

$$= \frac{(-1)^n}{(n+2)(n+1)}$$

\* each factor in  $n!$  is also in  $(n+2)!$

$$\lim_{n \rightarrow \infty} \frac{(-1)^n}{(n+2)(n+1)} = \boxed{0}$$

Patterns:

- Repeated addition  $\rightarrow C \cdot n$
- Repeated multiplication  $\rightarrow C^n$
- Factorial  $\rightarrow 1, 1, 2, 6, 24, 120, 720, \dots$
- Alternating  $\rightarrow (-1)^n$  or  $(-1)^{n+1}$
- $2^{n^2}$  common difference  $\rightarrow n^2$

Pattern Recognition for Sequences

**Ex. 3:** Find the  $n$ th term of the sequences described below.

a)  $\frac{4}{1}, \frac{8}{3}, \frac{16}{9}, \frac{32}{27}, \frac{64}{81}, \dots$

treat as 2 problems.  $\frac{\text{num}}{\text{den.}}$   
Pattern: repeated  $\cdot 2 \rightarrow 2^n$   
 repeated  $\cdot 3 \rightarrow 3^n$

Adjust: if only  $\frac{2^n}{3^n} = \frac{2}{3}, \frac{4}{9}, \frac{8}{27}, \frac{16}{81}, \dots$   
 So to make ours happen, multiply top by 2  
 one divide bottom by 3  $= \frac{2(2^n)}{3^n} = \frac{6 \cdot 2^n}{3^n}$

c)  $-1, 2, 7, 14, 23, \dots$

$+3 +5 +7 +9$   
 $+2 +2 +2$   $\leftarrow$  common 2<sup>nd</sup> difference

So  $n^2$   
 $n^2 = 1, 4, 9, 16, 25, \dots$  But  
 each term is 2 less, so

$n^2 - 2 = -1, 2, 7, 14, 23, \dots$

e)  $\frac{(x-2)}{6}, \frac{(x-2)^2}{12}, \frac{(x-2)^3}{24}, \frac{(x-2)^4}{48}, \dots$

$\frac{(x-2)^n}{2^n} \leftarrow$  repeated multiplication

$\frac{x-2}{2}, \frac{(x-2)^2}{4}, \frac{(x-2)^3}{8} \leftarrow$  good  $\div 3$

$\frac{(x-2)^n}{3 \cdot 2^n}$

b)  $-\frac{3}{1}, \frac{9}{2}, -\frac{27}{6}, \frac{81}{24}, -\frac{243}{120}, \dots$

Pattern: Alternating,  $\cdot 3$   
 Factorial

Alt:  $(-1)^n$  or  $(-1)^{n+1}$   
 if  $n=1$  is neg  $\rightarrow (-1)^n \checkmark$

$\frac{(-1)^n \cdot 3^n}{n!} = \frac{-3}{1}, \frac{9}{2}, \frac{-27}{6}, \dots$   
 no adjustments!

d)  $\frac{2}{1}, \frac{4}{3}, \frac{8}{5}, \frac{16}{7}, \frac{32}{9}, \dots$

$2^n \leftarrow$  "parent sequence"

$2n$   
Adjust:  $\frac{2}{2}, \frac{4}{4}, \frac{8}{6}, \frac{16}{8} \leftarrow$  good  
 $\frac{2^n}{2^{n-1}}$  subtract 1

f)  $1, -x, \frac{x^2}{2}, \frac{-x^3}{6}, \frac{x^4}{24}, \dots$

$\frac{1x^0}{1}, \frac{-1x^1}{1}, \frac{1x^2}{2}, \frac{-1x^3}{6}, \frac{1x^4}{24}, \dots$

$(-1)^n$  or  $(-1)^{n+1}$  ??  
 Starts at  $x^0$  and  $0!$ , so let's  
 begin with  $n=0$  instead of  $n=1$

$\frac{(-1)^n x^n}{n!} (n=0)$

$\frac{(-1)^{n+1} x^{n+1}}{(n+1)!}$  if  $n=1$

**AP Calculus II**  
**Notes 9.2**  
**Series and Convergence**

One important application of infinite sequences is the representation of infinite summations. The sum of an infinite sequence is an **infinite series**.

$$a_1 + a_2 + a_3 + \dots + a_n + \dots = \sum_{n=1}^{\infty} a_n$$

To find the sum of an infinite series (the limit), we examine the sequence of partial sums:

Consider the series:  $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n$



The sum of the first term is called the first partial sum. The second partial sum is the sum of the first 2 terms. The  $n$ th partial sum is the sum of the first  $n$  terms.

$$\sum_{n=1}^1 a_n = \frac{1}{2}$$

$$\sum_{n=1}^2 a_n = \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$$

$$\sum_{n=1}^3 a_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} = \frac{7}{8}$$

$$\sum_{n=1}^4 a_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} = \frac{15}{16}$$

$$\sum_{n=1}^5 a_n = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} = \frac{31}{32}$$

What is  $\sum_{n=1}^x a_n = ?$   $\frac{2^x - 1}{2^x}$

What is  $\lim_{x \rightarrow \infty} \sum_{n=1}^x a_n$  ?  $\lim_{x \rightarrow \infty} \frac{2^x - 1}{2^x} \rightarrow \frac{\infty}{\infty}$

$\lim_{x \rightarrow \infty} (2^x - 1) \Rightarrow \infty, \lim_{x \rightarrow \infty} 2^x \Rightarrow \infty \therefore \text{L'Hôpital}$   
 $= \lim_{x \rightarrow \infty} \frac{2^x \ln 2}{2^x \ln 2} = \boxed{1}$

**Definitions of Convergent and Divergent Series**

If the sequence of partial sums  $\{S_n\}$  converges to  $S$ , the infinite series  $\sum_{n=1}^{\infty} a_n$  converges and has a sum  $S$ . If  $\{S_n\}$  diverges, the series diverges.

**Ex. 1:** Determine if the sequence  $a_n$  converges and then determine if the series  $\sum_{n=1}^{\infty} a_n$  converges.

a)  $\sum_{n=1}^{\infty} \frac{1}{2} = \frac{1}{2} + \frac{1}{2} + \frac{1}{2} + \dots$

b)  $\sum_{n=0}^{\infty} 3(0.1)^n = 3 + 0.3 + 0.03 + 0.003 + \dots$

|          |                                                                                             |                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Sequence | $\lim_{n \rightarrow \infty} \left(\frac{1}{2}\right) = \frac{1}{2}$<br>so <b>converges</b> | $\lim_{n \rightarrow \infty} 3(0.1)^n \rightarrow 3(0) = 0$<br>each term <b>converges</b><br>will get closer to 0 |
| Series   | Always adding $\frac{1}{2}$ will<br>cause the sum to go<br>to $\infty$ , <b>Diverges</b>    | <u>Partial Sums:</u><br>$3, 3.3, 3.33, 3.333, \dots$<br><b>Converges to <math>3\frac{1}{3}</math></b>             |

**Geometric Series:**

$$\sum_{n=0}^{\infty} ar^n = a + ar + ar^2 + ar^3 + \dots + ar^n + \dots \quad a \neq 0 \quad \text{where } r \text{ is known as the}$$

Recall from Pre-Calculus that an infinite geometric series converges if  $|r| < 1$  and the sum of this convergent series is  $S_n = \frac{a_1}{1-r}$  where  $a_1$  is the first term. The series diverges if  $|r| \geq 1$ .

**Ex. 2:** Verify the convergence of the series from the previous example using the Geometric Series Test.

a)  $\sum_{n=1}^{\infty} \frac{1}{2}$

$= \sum_{n=1}^{\infty} \frac{1}{2}(1)^n$   
 which is geometric,  
 $|r| \geq 1$   
 $\therefore$  **Diverges**

b)  $\sum_{n=0}^{\infty} 3(0.1)^n$

$r = 0.1$   
 $|0.1| < 1$   
 $\therefore$  **converges**  
 $S = \frac{\text{first term}}{1-r}$   
 $= \frac{3(0.1)^0}{1-0.1} = \frac{3}{0.9} = \frac{10}{3}$

c)  $\sum_{n=0}^{\infty} \frac{8}{5^n} \leftarrow \text{not } \left(\frac{8}{5}\right)^n$

$= \sum_{n=0}^{\infty} 8\left(\frac{1}{5}\right)^n$   
 $|1/5| < 1$   
 $\therefore$  **converges**  
 $S = \frac{8}{1-1/5} \cdot \frac{5}{5}$   
 $= \frac{40}{5-1} = \frac{40}{4} = 10$

**Ex. 3:** Write the follow series in sigma notation. Then, determine whether the geometric series converges or diverges. If it converges, find the sum.

a)  $\frac{4}{3} + \frac{4}{9} + \frac{4}{27} + \frac{4}{81} + \dots = \sum_{n=1}^{\infty} \text{sequence}$  you can start at  $n=0, 1, 2 \dots$

$= \sum_{n=1}^{\infty} \frac{4}{3^n} = \sum_{n=1}^{\infty} 4\left(\frac{1}{3}\right)^n$   $|\frac{1}{3}| < 1$  so converges  
 $S = \frac{4/3}{1 - 1/3} \cdot \frac{3}{3} = \frac{4}{3-1} = \boxed{2}$

b)  $1 - \frac{e}{3} + \frac{e^2}{9} - \frac{e^3}{27} + \frac{e^4}{81} + \dots$  would be nice if start at  $n=0$   
 $= \sum_{n=0}^{\infty} \frac{e^n}{3^n} = \sum_{n=0}^{\infty} \left(\frac{e}{3}\right)^n$   $|\frac{e}{3}| < 1$  so converges  
 $S = \frac{1}{1 - e/3} \cdot \frac{3}{3} = \boxed{\frac{3}{3-e}}$

c)  $\frac{4}{3} + \frac{16}{9} + \frac{64}{27} + \frac{256}{81} + \dots$   
 $\sum_{n=1}^{\infty} \frac{4^n}{3^n} = \sum_{n=1}^{\infty} \left(\frac{4}{3}\right)^n$   $|\frac{4}{3}| \geq 1 \therefore$  Diverges

**Ex. 4:** Consider the geometric series  $\sum_{n=1}^{\infty} a_n$ , where  $a_n > 0$ , and  $\sum_{n=1}^{\infty} b_n$  for all  $n$ . The first term of both of the series is  $a_1 = b_1 = 720$ . The third term is  $a_3 = 80$  and the fourth term is  $b_4 = -90$ . Determine the convergence of both of the series, if they do.

$a_n$ :  $720, \dots, 80$  ,  $720r^2 = 80$  ,  $r^2 = \frac{1}{9}$  ,  $r = \frac{1}{3}$

$b_n$ :  $720, \dots, -90$  ,  $720r^3 = -90$  ,  $r^3 = -\frac{1}{8}$  ,  $r = -\frac{1}{2}$

$a_n = \sum_{n=0}^{\infty} 720\left(\frac{1}{3}\right)^n = \frac{720}{1 - 1/3} = \frac{2160}{3-1} = \boxed{1080}$   $|\frac{1}{3}| < 1$

$b_n = \sum_{n=0}^{\infty} 720\left(-\frac{1}{2}\right)^n = \frac{720}{1 + 1/2} = \frac{1440}{2+1} = \boxed{480}$   $|\frac{1}{2}| < 1$

## Properties of Infinite Series

Let  $\sum a_n = A$ ,  $\sum b_n = B$ , and  $c$  is a real number, then

1.  $\sum ca_n = cA$

2.  $\sum (a_n \pm b_n) = A \pm B$

**Ex. 5:** Find  $\sum_{n=1}^{\infty} \left[ 3(2)^{-2n} + \frac{2}{5^{n-1}} + \frac{(-4)^{n+1}}{5^n} \right]$  *split up!*

$$\sum_{n=1}^{\infty} \frac{3}{(2)^{2n}} = \sum_{n=1}^{\infty} \frac{3}{4^n} = \sum_{n=1}^{\infty} 3 \left( \frac{1}{4} \right)^n$$

$$S = \frac{3/4}{1 - 1/4} = \frac{3}{4-1} = 1$$

$$\sum_{n=1}^{\infty} 2 \left( \frac{1}{5} \right)^{n-1}$$

$$S = \frac{2 \cdot 5}{1 - 1/5} = \frac{10}{5-1} = 5/2$$

$$\sum_{n=1}^{\infty} \frac{(-4)^n \cdot 4}{5^n} = \sum_{n=1}^{\infty} 4 \left( \frac{-4}{5} \right)^n$$

$$S = \frac{-16}{5 - 4} = \frac{-16}{1-4} = \frac{-16}{-3} = \frac{16}{3}$$

$$= 1 + 5/2 - 16/9$$

**Ex. 6:** Find the values of  $x$  for which the following series converges. For these values, write the sum of the series as a function of  $x$ .

a)  $\sum_{n=1}^{\infty} \frac{x^n}{3^n} = \frac{x}{3} + \frac{x^2}{9} + \frac{x^3}{27} + \frac{x^4}{81} + \dots$

$$= \sum_{n=1}^{\infty} \left( \frac{x}{3} \right)^n \leftarrow \text{geometric}$$

will converge if

$$\left| \frac{x}{3} \right| < 1 \rightarrow -1 < \frac{x}{3} < 1$$

$$-3 < x < 3$$

$$S = \frac{x/3}{1 - x/3} = \frac{x}{3-x}$$

So for  $-3 < x < 3$ ,

$$\sum_{n=1}^{\infty} \left( \frac{x}{3} \right)^n = \frac{x}{3} + \frac{x^2}{9} + \frac{x^3}{27} + \dots = \frac{x}{3-x}$$

b)  $\sum_{n=0}^{\infty} 4 \left( \frac{x-3}{4} \right)^n \leftarrow \text{since geometric,}$

$$\left| \frac{x-3}{4} \right| < 1 \rightarrow -1 < \frac{x-3}{4} < 1$$

$$-4 < x-3 < 4 \rightarrow -1 < x < 7$$

$$S = \frac{4}{1 - (x-3)/4} = \frac{16}{4 - (x-3)}$$

$$S = \frac{16}{7-x}$$

Verify: if  $x=4$ ,  $\sum_{n=0}^{\infty} 4 \left( \frac{1}{4} \right)^n$

$$S = \frac{4}{1 - 1/4} = \frac{16}{4-1} = \frac{16}{3} \quad S(4) = \frac{16}{3}$$

## Nth Term Test for Divergence:

As a review, find the limit of each sequence and then determine the convergence of the following series:

|          |                                                  |                                                                                          |                                                                          |
|----------|--------------------------------------------------|------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|          | $\sum_{n=1}^{\infty} 2$                          | $\sum_{n=1}^{\infty} 2\left(\frac{4}{3}\right)^n$                                        | $\sum_{n=1}^{\infty} \frac{3n}{5n+4}$                                    |
| sequence | $\lim_{n \rightarrow \infty} 2 = 2$<br>converges | $\lim_{n \rightarrow \infty} 2\left(\frac{4}{3}\right)^n \rightarrow \infty$<br>Diverges | $\lim_{n \rightarrow \infty} \frac{3n}{5n+4} = \frac{3}{5}$<br>Converges |
| series   | $2+2+2+2+2+\dots$<br>Diverge                     | Since geometric<br>$ 4/3  \geq 1$ ,<br>Diverges                                          | Add up terms that<br>approach $3/5$ , it<br>will Diverge                 |

This is a small insight of the **nth Term Test for Divergence**, which states that:

For a series  $\sum_{n=1}^{\infty} a_n$ , if  $\lim_{n \rightarrow \infty} a_n \neq 0$ , then the series diverges.

### What this does and does not mean:

#### DOES MEAN

- if  $\lim_{n \rightarrow \infty} a_n \neq 0$ , the series Diverges
- if the series converges, then the  $\lim_{n \rightarrow \infty} a_n = 0$  ← "contrapositive"

#### DOES NOT MEAN

- if  $\lim_{n \rightarrow \infty} a_n = 0$ , the series converges  
→ FALSE! (it's actually inconclusive)
- if the series diverges, then  $\lim_{n \rightarrow \infty} a_n = 0$

THIS TEST CANNOT DETERMINE CONVERGENCE!!!!

↑  
which is why we say "Nth term test for Divergence"

**Ex. 7:** Use the nth term test for divergence, and any others learned, to determine if the series diverges:

a)  $\sum_{n=1}^{\infty} \left(\frac{6}{5}\right)^n \leftarrow \text{geometric!}$

$$|6/5| \geq 1$$

**Diverges**

b)  $\sum_{n=1}^{\infty} \frac{6}{5^n} = \sum_{n=1}^{\infty} 6\left(\frac{1}{5}\right)^n \leftarrow \text{geometric!}$

$$|1/5| < 1 \quad \text{Converges}$$

$$\text{to } S = \frac{6/5}{1 - 1/5} = \frac{6}{5-1} = \frac{3}{2}$$

c)  $\ln(1) + \ln\left(\frac{1}{2}\right) + \ln\left(\frac{1}{3}\right) + \ln\left(\frac{1}{4}\right) + \dots$

$$\sum_{n=1}^{\infty} \ln\left(\frac{1}{n}\right) \leftarrow \text{not geometric}$$

Nth term test for divergence:

$$\lim_{n \rightarrow \infty} \ln\left(\frac{1}{n}\right) \rightarrow \ln 0 \rightarrow -\infty \neq 0 \quad \therefore \text{Diverges}$$

e)  $\sum_{n=1}^{\infty} \frac{5n^2}{n^2 + 5n + 6}$

$$\lim_{n \rightarrow \infty} \frac{5n^2}{n^2 + 5n + 6} = 5 \neq 0$$

**Diverges**

d)  $\sum_{k=1}^{\infty} \frac{6^k}{k^2 + k + 1}$

$$\lim_{k \rightarrow \infty} \frac{6^k}{k^2 + k + 1} \quad \lim_{k \rightarrow \infty} 6^k \rightarrow \infty$$

$$\lim_{k \rightarrow \infty} \frac{6^k}{2k+1} \quad \lim_{k \rightarrow \infty} (k^2 + k + 1) \rightarrow \infty$$

$$\lim_{k \rightarrow \infty} \frac{6^k \ln 6}{2} \quad \lim_{k \rightarrow \infty} 2k+1 \rightarrow \infty$$

$$\lim_{k \rightarrow \infty} \frac{6^k \ln 6 \cdot \ln 6}{2} \rightarrow \infty \neq 0 \quad \therefore \text{Diverges}$$

f)  $\frac{1}{1} - \frac{2}{3} + \frac{4}{9} - \frac{8}{27} + \dots$

$$\frac{\text{repeated } \cdot 2}{\text{repeated } \cdot 3} \Rightarrow \left(\frac{-2}{3}\right)^n \leftarrow \text{good if } n=0 \text{ start}$$

$$\sum_{n=0}^{\infty} \left(\frac{-2}{3}\right)^n \rightarrow \frac{n^{\text{th}} \text{ term test for Divergence}}{\lim_{n \rightarrow \infty} \frac{(-2)^n}{3^n} \leftarrow \text{greater} = 0}$$

**INCONCLUSIVE!**

$$\sum_{n=0}^{\infty} \left(\frac{-2}{3}\right)^n \Rightarrow | -2/3 | < 1 \quad \therefore \text{Converges}$$

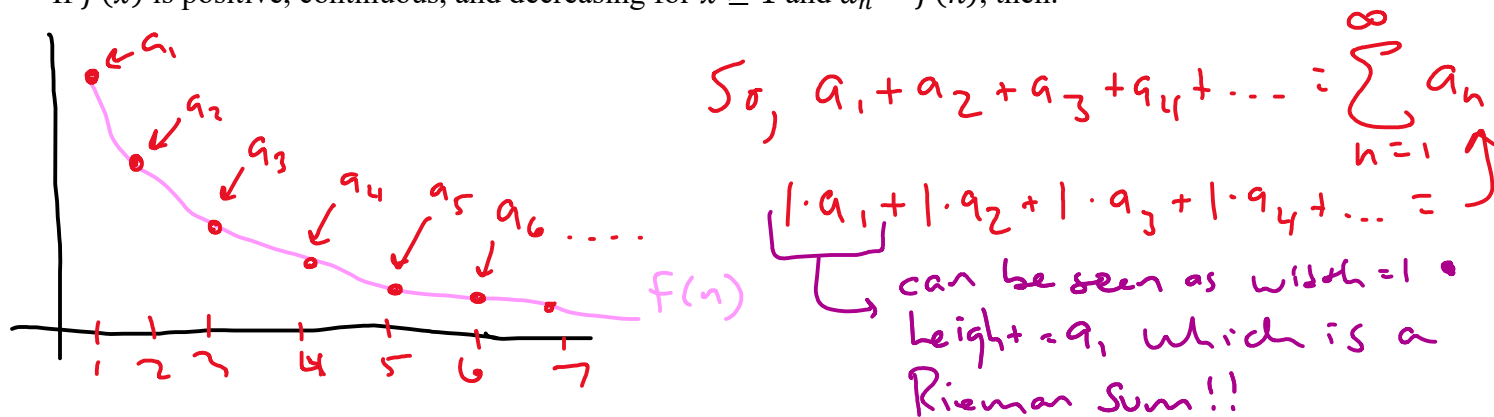
$$\text{to } S = \frac{1}{1 - (-2/3)} = \frac{3}{3+2} = \frac{3}{5}$$

AP Calculus II  
Notes 9.3  
Integral Test and p-Series

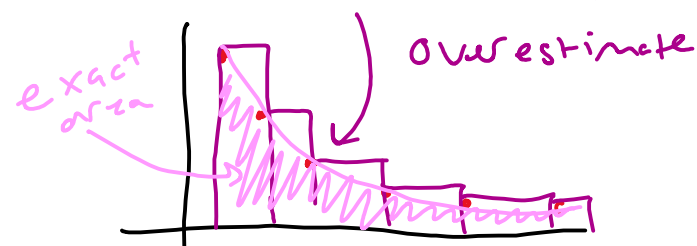
Over the next two sections, we will learn several convergence tests that apply to series with *positive* terms.

**The Integral Test**

If  $f(x)$  is positive, continuous, and decreasing for  $x \geq 1$  and  $a_n = f(n)$ , then:

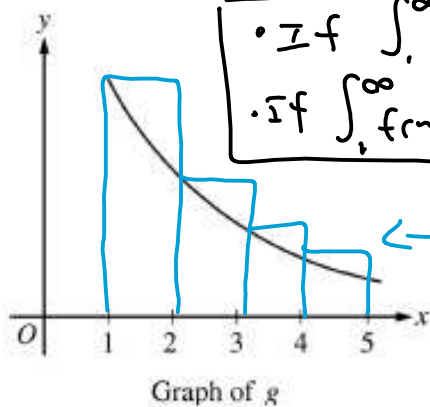


Left Riemann Sum



So, Left Riemann =  $\sum_{n=1}^{\infty} a_n$   
 $\sum_{n=1}^{\infty} a_n > \int_1^{\infty} f(x) dx$

Ex. 1:



• If  $\int_1^{\infty} f(x) dx$  diverges, the  $\sum a_n$  diverges  
 • If  $\int_1^{\infty} f(x) dx$  converges, the  $\sum a_n$  converges

(A)  $\sum_{n=1}^{\infty} a_n = 6$

(B)  $\sum_{n=1}^{\infty} a_n = 8$

(C)  $\sum_{n=1}^{\infty} a_n = 10$

(D)  $\sum_{n=1}^{\infty} a_n$  diverges

For  $x \geq 1$ , the continuous function  $g$  is decreasing and positive. A portion of the graph of  $g$  is shown above. For  $n \geq 1$ , the  $n$ th term of the series  $\sum_{n=1}^{\infty} a_n$  is defined by  $a_n = g(n)$ . If

$\int_1^{\infty} g(x) dx$  converges to 8, which of the following could be true?

if  $\int_1^{\infty} g(x) dx = 8$  and converges, then  $\sum_{n=1}^{\infty} a_n$  converges

**Ex. 2:** Determine the convergence of the following series:

a)  $\sum_{n=1}^{\infty} \frac{n}{n^2+1}$  ← not geometric... maybe  $n^{\text{th}}$  term test for divergence?

$\lim_{n \rightarrow \infty} \frac{n}{n^2+1} = 0$  ← inconclusive!

Integral Test: Since  $f$  is positive, continuous, decreasing

$u = n^2+1, du = 2n dn$

$\int_1^{\infty} \frac{n}{n^2+1} dn$

$\lim_{b \rightarrow \infty} \frac{1}{2} \ln|n^2+1| \Big|_1^b$

$\lim_{b \rightarrow \infty} \left[ \frac{1}{2} \ln|b^2+1| - \frac{1}{2} \ln 2 \right]$

as  $b \rightarrow \infty, \ln|b^2+1| \rightarrow \ln \infty \rightarrow \infty$

So,  $\int_1^{\infty} \frac{n}{n^2+1} dn$  Diverges, so  $\sum_{n=1}^{\infty} \frac{n}{n^2+1}$  **Diverges**

p-Series and Harmonic Series

The second test for convergence of a series has a simple arithmetic form:

$\sum_{n=1}^{\infty} \frac{1}{n^p} = \frac{1}{n} + \frac{1}{n^2} + \frac{1}{n^3} + \frac{1}{n^4} + \dots$  (Note this is different than Geometric!)

which is known as a **p-series**, where  $p$  is a positive constant. For  $p = 1$ , the series:

$\sum_{n=1}^{\infty} \frac{1}{n} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \dots$  ← very famous

is known as the **harmonic series**. The Greek used this in music to create the length of strings of the same material, diameter, etc... that produce harmonic tones.

$p$  is meant to be positive power when in denominator

b)  $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$  Since  $f$  is positive, continuous & decreasing,

$\int_1^{\infty} \frac{1}{n^2+1} dn$  arctan!

$\lim_{b \rightarrow \infty} \arctan n \Big|_1^b$

$\lim_{b \rightarrow \infty} [\arctan b - \arctan 1]$   
as  $b \rightarrow \infty, \arctan b \rightarrow \boxed{\pi/2}$

$\pi/2 - \pi/4 = \pi/4$

So,  $\int_1^{\infty} \frac{1}{n^2+1} dn$  converges to  $\pi/4$

and  $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$  converges  
BUT NOT TO  $\pi/4$ !!  
(just converges)

The Integral Test is useful in determining the convergence and divergence of this  $p$ -series.

**Ex. 3:** Apply the Integral Test to determine the  $p$ -series rules:

a)  $\sum_{n=1}^{\infty} \frac{8}{\sqrt{n}} = \sum_{n=1}^{\infty} \frac{8}{n^{1/2}}$

Since  $f(n) = \frac{8}{\sqrt{n}}$  is positive, continuous and decreasing (for  $n \geq 1$ )

$$\int_1^{\infty} \frac{8}{n^{1/2}} dn = \lim_{b \rightarrow \infty} 8 \int_1^b n^{-1/2} dn$$

$$\lim_{b \rightarrow \infty} 16\sqrt{n} \Big|_1^b = \lim_{b \rightarrow \infty} [16\sqrt{b} - 16]$$

goes to  $\infty$

$\therefore$  **Diverges** ( $p = 1/2$ )

b)  $5 + \frac{5}{4} + \frac{5}{9} + \frac{5}{16} + \dots = \sum_{n=1}^{\infty} \frac{5}{n^2}$

Since  $f(n) = \frac{5}{n^2}$  is positive, continuous ( $n \geq 1$ ) and decreasing,

$$\int_1^{\infty} \frac{5}{n^2} dn = \lim_{b \rightarrow \infty} 5 \int_1^b n^{-2} dn$$

$$\lim_{b \rightarrow \infty} \frac{-5}{n} \Big|_1^b = \lim_{b \rightarrow \infty} \left[ \frac{-5}{b} - \frac{-5}{1} \right]$$

goes to 0

$\therefore$  **Converges** ( $p = 2$ )

Convergence of  $p$ -Series proof:

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \sum_{n=1}^{\infty} n^{-p}$$

since  $f(n) = \frac{1}{n^p}$  is positive, decreasing and continuous

$$\int_1^{\infty} n^{-p} dn \rightarrow \lim_{b \rightarrow \infty} \int_1^b n^{-p} dn = \lim_{b \rightarrow \infty} \frac{n^{-p+1}}{-p+1} \Big|_1^b =$$

$$\lim_{b \rightarrow \infty} \frac{1}{1-p} \cdot n^{1-p} \Big|_1^b = \lim_{b \rightarrow \infty} \left[ \frac{1}{1-p} \cdot b^{1-p} - \frac{1}{1-p} \cdot (1)^{1-p} \right]$$

$1-p$  is negative, if  $\infty^{1-p}$   $1-p$  is positive

Converges:  $1-p < 0, 1 < p$  Diverges

a) converges if  **$p > 1$**

b) diverges if  **$p \leq 1$**

Note that for a  $p$ -series, the  $p$ -value is considered positive when in the denominator and could possibly have a constant multiple. This may require a little rewriting or simplification to get a series in this form.

opposite of geometric, so careful!

**Ex. 4:** Determine if the following series converge or diverge:

a)  $\sum_{k=1}^{\infty} \frac{4}{k^{1.001}}$

$p = 1.001 \geq 1$

**Converges**

b)  $\sum_{n=1}^{\infty} \frac{n^2}{3}$

$n^{\text{th}}$  term test for Divergence

$\lim_{n \rightarrow \infty} \frac{n^2}{3} = \infty \neq 0$

$\therefore$  Diverges

c)  $1 + \frac{1}{\sqrt[3]{2}} + \frac{1}{\sqrt[3]{3}} + \frac{1}{\sqrt[3]{4}} + \dots$

$= \sum_{n=1}^{\infty} \frac{1}{\sqrt[3]{n}}$

p-series:  $p = 1/3 \leq 1$

**Diverges**

d)  $\frac{4}{205} + \frac{5}{207} + \frac{6}{209} + \frac{7}{211} + \frac{8}{213} + \dots$

$\sum_{n=0}^{\infty} \frac{n+4}{2n+205}$

$n^{\text{th}}$  term test for Divergence

$\lim_{n \rightarrow \infty} \frac{n+4}{2n+205} = \frac{1}{2} \neq 0$

**Diverges**

e)  $\frac{6}{2} + \frac{6}{5} + \frac{6}{10} + \frac{6}{17} + \frac{6}{26} + \dots$

$+3, +5, +7, +9$   
 $+2, +2, +2$   $\swarrow$  2<sup>nd</sup> common difference

$\sum_{n=1}^{\infty} \frac{6}{n^2+1}$

$+1$  means it's not p-series!

Since  $f(n) = \frac{6}{n^2+1}$  is positive, continuous and decreasing

$\int_1^{\infty} \frac{6}{n^2+1} dn = \lim_{b \rightarrow \infty} \int_1^b \frac{6}{n^2+1} dn$

$\lim_{b \rightarrow \infty} \left[ 6 \arctan n \right]_1^b = \lim_{b \rightarrow \infty} [6 \arctan b - 6 \arctan 1]$   
 $= 6 \cdot \pi/2 - 6 \cdot \pi/4$   
 $\therefore$  Converges

f)  $\sum_{n=1}^{\infty} \left(\frac{4}{5}\right)^{n-1}$

geometric!

$\left|\frac{4}{5}\right| < 1$

$\therefore$  Converges

$S = \frac{1}{1 - 4/5} \cdot \frac{5}{5}$

$= \frac{5}{5-4} = 5$

# AP Calculus II

## Notes 9.4

### Comparisons of Series

For all the convergence tests that we have used so far, the series have been very simple and have fit into each of the tests exactly. A small deviation from these special characteristics can make a test inapplicable. These tests are very similar to integration rules. For example, check out the following pairs:

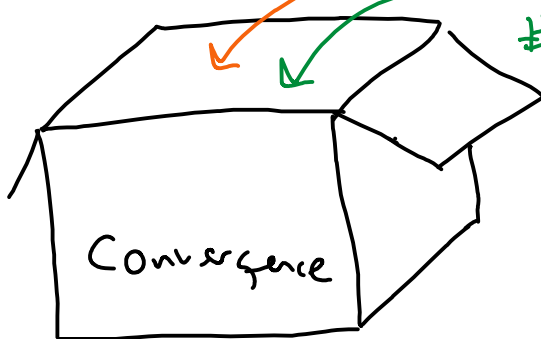
a)  $\sum_{n=0}^{\infty} \frac{1}{2^n}$  is *geometric*    b)  $\sum_{n=1}^{\infty} \frac{1}{n^3}$  is *p-series*    c)  $\sum_{n=1}^{\infty} \frac{n}{(n^2+1)^2}$  is *integral test*

$\sum_{n=0}^{\infty} \frac{n}{2^n}$  is *NOT geometric*     $\sum_{n=1}^{\infty} \frac{1}{n^3+1}$  is *not p-series*     $\sum_{n=1}^{\infty} \frac{n^2}{(n^2+1)^2}$  is *not integral*

In this section we will learn two additional tests for positive term series. These two tests will allow us to determine the convergence or divergence of a whole new set of series. They will allow us to **compare** a series having complicated terms with a **simpler** series whose convergence or divergence is known.

#### Visual Representation of Comparison Tests

Box analogy!



#2 fits in box



#1

#2

#3

• if you can close the top, the series converges (adds to number)

#### The Direct Comparison Test

Let  $0 \leq a_n \leq b_n$  for all  $n$ .

1) If  $\sum_{n=1}^{\infty} b_n$  converges,  $\sum a_n$  converges

2) If  $\sum_{n=1}^{\infty} b_n$  diverges,  $\sum a_n$  inconclusive

3) If  $\sum_{n=1}^{\infty} a_n$  diverges,  $\sum b_n$  diverges

4) If  $\sum_{n=1}^{\infty} a_n$  converges,  $\sum b_n$  inconclusive

1) If we know #2 converges, then smaller #1 must converge. #3 is unknown

2) If we know #2 diverges, then larger #3 diverges, #1 is unknown

Pick a series whose convergence/divergence is known!

**Ex. 1:** Determine the convergence or divergence of the following series:

a)  $\sum_{n=0}^{\infty} \frac{1}{2+3^n}$   $\leftarrow$  looks like  $\sum \frac{1}{3^n}$

\* since  $\sum \frac{1}{3^n}$  is convergence, we know it "fits in the box". So, if we can show  $\sum \frac{1}{2+3^n}$  is smaller, that it must also converge!

Show:  $\sum \frac{1}{2+3^n} < \sum \frac{1}{3^n}$ . will be true if  $\frac{1}{2+3^n} < \frac{1}{3^n} \rightarrow 3^n < 2+3^n$  true!!

b)  $\sum_{k=0}^{\infty} \frac{3}{4+\sqrt{k}}$   $\leftarrow \therefore \sum \frac{1}{2+3^n}$  Converges

Compare to  $\sum \frac{3}{\sqrt{k}}$ , which is divergent  $p = 1/2 \leq 1$

So, we must show  $\sum \frac{3}{4+\sqrt{k}}$  is greater than  $\sum \frac{3}{\sqrt{k}}$  since being bigger than divergent is also divergent

$\frac{3}{4+\sqrt{k}} \geq \frac{3}{\sqrt{k}} \rightarrow 3\sqrt{k} \geq 3(4+\sqrt{k}) \leftarrow \text{FALSE}$

So we don't know.... yet....

c)  $\sum_{k=2}^{\infty} \frac{\ln k}{k^4+1}$  \*  $\ln k$  always less than  $k$  \*

Compare to  $\sum \frac{k}{k^4+1}$ , which can be compared to  $\sum \left( \frac{k}{k^4} = \frac{1}{k^3} \right)$

This is a convergent p-series, so we need the original series to be less.

Is  $\frac{1}{k^3} > \frac{k}{k^4+1}$ ?  $\rightarrow k^4+1 > k^4 \leftarrow \text{TRUE!}$  So,  $\sum \frac{k}{k^4+1}$  is also convergent.

Is  $\frac{k}{k^4+1} > \frac{\ln k}{k^4+1}$ ? Yes!  $k > \ln k \rightarrow$  So  $\sum \frac{\ln k}{k^4+1}$  converges

Sometimes, we can see that a series resembles a p-series or a geometric series, but we can't establish the necessary comparison that satisfies the conditions of the Direct Comparison Test. When this happens, you may be able to apply the second comparison test known as the **Limit Comparison Test**.

### Limit Comparison Test

Suppose that  $a_n > 0, b_n > 0$ , and  $\lim_{n \rightarrow \infty} \left( \frac{a_n}{b_n} \right) = L$  where  $L$  is **finite** and **positive**. This ratio will say that as the terms go on to infinity, they are off by a factor of a constant multiple. Therefore, the sum of these terms will be off by that constant multiple (about). Then, the two series either both converge or both diverge.

**Ex. 2:** Determine the convergence or divergence of the following series:

a)  $\sum_{n=1}^{\infty} \frac{4}{3n^2 - 2n + 5}$  *strip down all constants and lower functions!*

Compare to  $\sum \frac{1}{n^2}$

$$\lim_{n \rightarrow \infty} \left( \frac{\frac{4}{3n^2 - 2n + 5}}{\frac{1}{n^2}} \right) = \lim_{n \rightarrow \infty} \left( \frac{4n^2}{3n^2 - 2n + 5} \right)$$

$= \boxed{4/3}$ . So, this means, each term in  $\frac{4}{3n^2 - 2n + 5}$  is about  $\frac{4}{3}$  each term in  $\frac{1}{n^2}$ . So, since  $\sum \frac{1}{n^2}$  converges,  $\sum \frac{4}{3n^2 - 2n + 5}$  must also

c)  $\sum_{k=1}^{\infty} \frac{2\sqrt[3]{5}k^4}{(7k+1)^2}$

compare to  $\sum \frac{k^{4/3}}{k^2} = \sum \frac{1}{k^{2/3}}$

$$\lim_{k \rightarrow \infty} \left( \frac{\frac{2\sqrt[3]{5}k^4}{(7k+1)^2}}{\frac{1}{k^{2/3}}} \right) = \lim_{k \rightarrow \infty} \left( \frac{2\sqrt[3]{5}k^2}{(7k+1)^2} \right)$$

$= \frac{2\sqrt[3]{5}}{49}$ . So since  $\sum \frac{1}{k^{2/3}}$  is a divergent p-series, this **Diverges**

b)  $\sum_{n=1}^{\infty} \frac{3^n}{5^n - 2}$

compare to  $\sum \frac{3^n}{5^n}$

$$\lim_{n \rightarrow \infty} \left( \frac{\frac{3^n}{5^n - 2}}{\frac{3^n}{5^n}} \right) = \lim_{n \rightarrow \infty} \left( \frac{15^n}{15^n \cdot 2 \cdot 3^n} \right)$$

*insignificant*

$50 = 1$ . Therefore, since  $\sum \left(\frac{3}{5}\right)^n$  converges, So  $\sum \frac{3^n}{5^n - 2}$  also converges

d)  $\sum_{n=1}^{\infty} \frac{6n^8 + 5n^7 - 3n^2 + 2}{n^3(2n+5)^3(3n^3-2)}$

Compare to  $\sum \frac{n^8}{n^3 \cdot n^3 \cdot n^3}$

$$= \sum \frac{1}{n}$$

$$\lim_{n \rightarrow \infty} \left( \frac{\frac{6n^8 + \dots}{n^3(2n+5)^3(3n^3-2)}}{\frac{1}{n}} \right) = L$$

So **Diverges** (harmonic!)

**Ex. 3:** Determine the convergence or divergence of the following series using any technique learned so far:

a)  $\sum_{n=1}^{\infty} \frac{6n^2 + 17}{2n^5 - n^3 + 5n}$

Compare to  $\sum \frac{n^2}{n^5} = \sum \frac{1}{n^3}$   
 $p=3 > 1$

**Converges**

b)  $\sum_{k=0}^{\infty} \frac{2^k}{\pi^k + 7}$

compare to  $\sum \left(\frac{2}{\pi}\right)^k$

$\left|\frac{2}{\pi}\right| < 1 \therefore$  **converges**

c)  $\sum_{n=1}^{\infty} \frac{n^2}{\sqrt{n} + 3}$

$\lim_{n \rightarrow \infty} \frac{n^2}{\sqrt{n} + 3} = \infty \neq 0$

$\therefore$  **Diverges**

d)  $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2 + 3}$

compare to  $\sum \frac{n^{1/2}}{n^2} = \sum \frac{1}{n^{3/2}}$

$p = 3/2 > 1$

**Converges**

e)  $\sum_{n=1}^{\infty} \frac{2n}{(n^2 + 5)^3}$

Compare to  $\sum \frac{1}{n^5}$

**converges**

$p=5 > 1$

Which of the following series cannot be shown to converge using the limit comparison test with the series  $\sum_{n=1}^{\infty} \frac{1}{n^2}$ ?

(A)  $\sum_{n=1}^{\infty} \frac{4}{3n^2 - n}$  ✓

(B)  $\sum_{n=1}^{\infty} \frac{15}{\sqrt{n^4 + 5}}$  ✓

(C)  $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n^2}\right)$  ?

(D)  $\sum_{n=1}^{\infty} \frac{\ln n}{n^2}$   $\frac{n}{n^2} = \frac{1}{n}$

**AP Calculus II**  
**Notes 9.5**  
**Alternating Series**

So far, the majority of the series we have dealt with have had positive terms. Now, we will look at series that have both positive and negative terms. This is known as an **alternating series**.

**Alternating Series Test:**

Let  $a_n > 0$ . The alternating series

$$\sum_{n=1}^{\infty} (-1)^n a_n \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^{n+1} a_n \quad \text{or} \quad \sum_{n=1}^{\infty} (-1)^{n-1} a_n$$

converge iff **both** of the following conditions are met:

1.  $\lim_{n \rightarrow \infty} a_n = 0$

2.  $a_{n+1} < a_n$   
 ↑  
 next term < previous term

**Demonstration:** Determine the convergence of the following using the Geometric and Alternating series.

$$\sum_{n=0}^{\infty} \left(-\frac{1}{2}\right)^n = \sum_{n=0}^{\infty} (-1)^n \left(\frac{1}{2}\right)^n =$$

$$\sum_{n=4}^{\infty} \left(-\frac{5}{4}\right)^{n-3} = \sum_{n=4}^{\infty} (-1)^{n-3} \left(\frac{5}{4}\right)^{n-3} =$$

i)  $\lim_{n \rightarrow \infty} \left(\frac{1}{2}\right)^n = 0 \checkmark$

ii)  $\left(\frac{1}{2}\right)^{n+1} < \left(\frac{1}{2}\right)^n \checkmark \checkmark$

∴ converges

Also,  $|-1/2| < 1 \rightarrow$  converges

i)  $\lim_{n \rightarrow \infty} \left(\frac{5}{4}\right)^{n-3} = \infty \neq 0$

Fails Alternating Series

$|5/4| \geq 1 \therefore$  Diverges by Geometric

Note that sometimes alternating series are hidden or appear to be alternating but are not. For example,

$$\sum_{n=1}^{\infty} (-1)^{2n} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} (-1)^{2n+1} a_n \quad \text{aren't alternating}$$

while  $\sum_{n=1}^{\infty} (-1)^{3n-1} a_n \quad \text{and} \quad \sum_{n=1}^{\infty} \cos(\pi n) a_n \quad \text{are alternating}.$

**THIS TEST CANNOT DETERMINE DIVERGENCE!!!!**

**Ex. 1:** Determine the convergence or divergence of the following series:

a)  $\sum_{n=1}^{\infty} (-1)^n \frac{11}{n+4}$

i)  $\lim_{n \rightarrow \infty} \frac{11}{n+4} = 0 \quad \checkmark$

ii)  $\frac{11}{(n+1)+4} < \frac{11}{n+4}$

larger denominator  $\rightarrow \frac{11}{n+5} < \frac{11}{n+4} \quad \checkmark \checkmark$

$\therefore$  **Converges**

b)  $-\frac{4}{1} + \frac{5}{2} - \frac{6}{6} + \frac{7}{24} - \frac{8}{120} + \frac{9}{720} - \dots$

$\sum_{n=1}^{\infty} \frac{(-1)^n (n+3)}{n!}$

i)  $\lim_{n \rightarrow \infty} \frac{n+3}{n!} = 0 \quad \checkmark$   
gets to  $\infty$  very fast

ii)  $\frac{n+4}{(n+1)!} < \frac{n+3}{n!} \quad \checkmark \checkmark$

Which is more significant: adding 1 to numerator or having bigger factorial? Factorial so  $\checkmark$

$\therefore$  **Converges**

c)  $\sum_{x=2}^{\infty} (-1)^{x+1} \frac{x+1}{\ln x}$

i)  $\lim_{x \rightarrow \infty} \frac{x+1}{\ln x} = 0?$

$\lim_{x \rightarrow \infty} (x+1) = \infty, \lim_{x \rightarrow \infty} \ln x = \infty \therefore$  L'Hôp

$= \lim_{x \rightarrow \infty} \frac{1}{\frac{1}{x}} = \lim_{x \rightarrow \infty} x = \infty \neq 0$

Fails Alternating Series.

BUT,  $n^{\text{th}}$  term test for

divergence says **Diverges**

d)  $\sum_{k=1}^{\infty} \frac{k}{(-3)^{k-1}} = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{k}{3^k}$

i)  $\lim_{k \rightarrow \infty} \frac{k}{3^k} = 0?$

$\lim_{k \rightarrow \infty} k = \infty, \lim_{k \rightarrow \infty} 3^k = \infty \therefore$  L'Hôp

$= \lim_{k \rightarrow \infty} \frac{1}{3 \cdot \ln 3} = 0 \quad \checkmark$

ii)  $\frac{k+1}{3^{k+1}} < \frac{k}{3^k}?$

Add 1 to power is more significant, so  $a_{k+1}$  is smaller!  $\checkmark \checkmark$

$\therefore$  **Converges**

## Absolute and Conditional Convergence

### Conditional Convergence

If  $\sum_{n=1}^{\infty} (-1)^n a_n$  converges but  $\sum_{n=1}^{\infty} |a_n|$  diverges, then  $\sum_{n=1}^{\infty} (-1)^n a_n$  is **conditionally convergent**.

Reason for Conditional:  $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt[3]{n}}$

Alternating Series:

i)  $\lim_{n \rightarrow \infty} \frac{1}{\sqrt[3]{n}} = 0 \checkmark$

ii)  $\frac{1}{\sqrt[3]{n+1}} < \frac{1}{\sqrt[3]{n}} \checkmark \checkmark$

... Converges

What if it wasn't  
alternating??

$\sum \frac{1}{n^{1/3}} \quad p = 1/3 \leq 1$

$\therefore$  **Diverges**

So, only reason  $\sum \frac{(-1)^n}{\sqrt[3]{n}}$   
converges is because it's alternating  
So **Conditional convergence**

### Absolute Convergence

If  $\sum_{n=1}^{\infty} (-1)^n a_n$  converges and  $\sum_{n=1}^{\infty} |a_n|$  converges, then  $\sum_{n=1}^{\infty} (-1)^n a_n$  is **absolutely convergent**.

What is an alternating series that is absolutely convergent?

$$\sum \frac{(-1)^n}{n^2} \rightarrow \sum \left| \frac{(-1)^n}{n^2} \right| = \sum \frac{1}{n^2} \text{ which is } p=2 > 1$$

Convergent.

i)  $\lim_{n \rightarrow \infty} \frac{1}{n^2} = 0 \checkmark$

ii)  $\frac{1}{(n+1)^2} < \frac{1}{n^2} ? \checkmark \checkmark$

$\therefore$  **Absolute convergence**

How could we use the Direct Comparison Test to further support the concept of absolute convergence?

Since all positive series converges, the smaller  
(alternating) series must also converge by  
the Direct Comparison Test.

**Ex. 2:** Determine the convergence or divergence of the following series. Classify any convergent series as absolutely or conditionally convergent.

a)  $\sum_{n=1}^{\infty} (-1)^n \frac{n!}{2^n}$

i)  $\lim_{n \rightarrow \infty} \frac{n!}{2^n} = 0?$  NO

So  $n^{\text{th}}$  term test for

Divergence means

**Diverges**

b)  $\sum_{n=0}^{\infty} \frac{\cos(n\pi)}{\sqrt{n+4}}$  ← alternating

i)  $\lim_{n \rightarrow \infty} \frac{1}{\sqrt{n+4}} = 0?$  Yes ✓

ii)  $\frac{1}{\sqrt{n+5}} < \frac{1}{\sqrt{n+4}}?$  Yes ✓

∴ **Converges**

$\sum \frac{1}{\sqrt{n+4}}$  compare to  $\sum \frac{1}{\sqrt{n}}$  ← diverges  
∴ **Conditional**

c)  $\sum_{n=1}^{\infty} (-1)^{n-1} \frac{\sqrt[3]{n}}{\sqrt{n}} = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{1}{n^{1/6}}$

$\frac{n^{1/3}}{n^{1/2}} = \frac{1}{n^{1/6}}$

i)  $\lim_{n \rightarrow \infty} \frac{1}{n^{1/6}} = 0?$  ✓

ii)  $\frac{1}{(n+1)^{1/6}} < \frac{1}{n^{1/6}}?$  ✓✓

converges!

$\sum \frac{1}{n^{1/6}}$  diverges  $p = 1/6 \leq 1$

∴ **Conditional converges**

e) Which of the following series is absolutely convergent?

$\sum \frac{1}{2n} \rightarrow$  (A)  $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{2n}$

all positive converges

$\sum \frac{1}{\sqrt{n}} \rightarrow$  (B)  $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{1}{\sqrt{n}}$

harmonic p-series

$\sum \frac{n}{n+1} \rightarrow$  (C)  $\sum_{n=1}^{\infty} (-1)^{n+1} \frac{n}{n+1}$

$n^{\text{th}}$  term test for divergence

$\sum \left(\frac{1}{2}\right)^n \rightarrow$  (D)  $\sum_{n=1}^{\infty} (-1)^{n+1} \left(\frac{1}{2}\right)^n$

↑ geometric!

d)  $\sum_{n=2}^{\infty} \frac{(-1)^n}{n \ln n}$

i)  $\lim_{n \rightarrow \infty} \frac{1}{n \ln n} = 0$  ✓

ii)  $\frac{1}{(n+1) \ln(n+1)} < \frac{1}{n \ln n}?$  ✓✓

$\sum \frac{1}{n \ln n} \rightarrow \int_2^{\infty} \frac{1}{u \ln u} du$  Since  $f$  is positive, continuous, decreasing

$\lim_{b \rightarrow \infty} \int_2^b \frac{1}{u} du = \ln|u| \Big|_2^b = \ln b - \ln 2 = \infty$   $u = \ln u, du = \frac{1}{u} du$

$\lim_{b \rightarrow \infty} \ln|b| - \ln|2| = \infty$

∴ **cond. Converge**

f)  $\frac{5}{1} - \frac{6}{8} + \frac{7}{27} - \frac{8}{64} + \frac{9}{125} - \dots$

$\sum_{n=1}^{\infty} \frac{(-1)^{n+1} (n+4)}{n^3}$

i)  $\lim_{n \rightarrow \infty} \frac{n+4}{n^3} = 0$  ✓

ii)  $\frac{n+5}{(n+1)^3} < \frac{n+4}{n^3}?$  ✓✓

$\sum \frac{n+4}{n^3}$  compare to  $\sum \frac{1}{n^2}$  convergent

∴ **Absolute convergence**

**AP Calculus II**  
**Notes 9.6**  
**Ratio Tests**

This section begins with a test for absolute convergence known as the **Ratio Test**.

**Ratio Test**

Let  $\sum_{n=1}^{\infty} a_n$  be a series with nonzero terms.

1.  $\sum_{n=1}^{\infty} a_n$  converges absolutely if  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$ .
2.  $\sum_{n=1}^{\infty} a_n$  diverges if  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| > 1$  or  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \infty$ .
3. The Ratio Test is inconclusive if  $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$ .

Connection to Geometric Series and Inconclusive Examples:

Definition of  $r$  in Geometric is  $\frac{\text{next term}}{\text{previous term}} = r$

which is what we find here

Geometric:  $|r| < 1 \rightarrow \text{converges}$   
 $|r| > 1 \rightarrow \text{diverges}$

But, Algebraic expressions give  $r=1$ , but some algebraic  $\frac{1}{\sqrt{n}}$  vs  $\frac{1}{n^4}$  have conflicting results.  
So inconclusive

The Ratio Test is not a test that works for everything, but it is very useful for series that converge rapidly. These include factorials and exponentials. It especially is useful when a series has multiple parts. It also does not matter whether the series alternates or does not.

**Ex. 1:** Determine the convergence and divergence of the following series:

a)  $\sum_{n=1}^{\infty} (-1)^{n-1} n \left(\frac{10}{9}\right)^{n+1}$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1) \left(\frac{10}{9}\right)^{n+2}}{n \left(\frac{10}{9}\right)^{n+1}} \right|$$

$$= \lim_{n \rightarrow \infty} \left| \frac{10/9 (n+1)}{n} \right| = \frac{10}{9} > 1$$

$\therefore$  **Diverges**

Alternating  
disappears  
with absolute  
value

b)  $\sum_{k=1}^{\infty} (-1)^{2k+1} \frac{4^k}{k!}$

$$\lim_{k \rightarrow \infty} \left| \frac{4^{k+1}}{(k+1)!} \cdot \frac{k!}{4^k} \right| =$$

$$\lim_{k \rightarrow \infty} \left| \frac{4^{k+1} \cdot k!}{(k+1)! \cdot 4^k} \right| \rightarrow \left| \frac{4}{k+1} \right|$$

$$\lim_{k \rightarrow \infty} \left| \frac{4}{k+1} \right| = 0 < 1$$

$\therefore$  **Converges**

c)  $\sum_{n=2}^{\infty} \frac{5n^2 2^{n+1}}{3^{2n}}$

$$\lim_{n \rightarrow \infty} \left| \frac{5(n+1)^2 \cdot 2^{n+2}}{3^{2n+2}} \cdot \frac{3^{2n}}{5n^2 \cdot 2^{n+1}} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{2(n+1)^2}{9n^2} \right| = \frac{2}{9} < 1$$

$\therefore$  **Converges**

d)  $\sum_{n=1}^{\infty} \frac{(-1)^n (n+2)}{n(n+1)}$

$$\lim_{n \rightarrow \infty} \left| \frac{n+3}{(n+1)(n+2)} \cdot \frac{n(n+1)}{n+2} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n(n+3)}{(n+2)^2} \right| = 1 \text{ (inclusive)}$$

makes sense because there are  
factorials or geometric parts...

Alternating!

i)  $\lim_{n \rightarrow \infty} \frac{n+2}{n(n+1)} = 0$  ? ✓

ii)  $\frac{n+3}{(n+1)(n+2)} < \frac{n+2}{n(n+1)}$  ? ✓✓

$\therefore$  **Converges**

**Ex. 2:**

BC3: Let  $f(x) = -\frac{2}{\sqrt{x}}$  and let  $A = \sum_{n=1}^{\infty} a_n$  where  $a_n = f(n)$ .

(a) Determine whether  $\sum_{n=1}^{\infty} a_n$  and  $\sum_{n=1}^{\infty} (-1)^{n-1} a_n$  converge or diverge. Explain your reasoning.

not alternating!  $\sum \frac{-2}{\sqrt{n}}$   $p = 1/2 \leq 1$  Diverges  $\left\{ \begin{array}{l} \sum (-1)^{n-1} \cdot \frac{-2}{\sqrt{n}} \text{ i) } \lim_{n \rightarrow \infty} \frac{2}{\sqrt{n}} = 0? \checkmark \\ \text{ii) } \frac{2}{\sqrt{n+1}} < \frac{2}{\sqrt{n}}? \checkmark \checkmark \end{array} \right.$

(b) Consider  $\sum_{n=1}^{\infty} b_n$  where  $b_n = f'(n)$ . Determine whether  $\sum_{n=1}^{\infty} b_n$  converges or diverges. converges by alternating series

Explain your reasoning.

$f' = -2 \cdot -1/2 x^{-3/2}$   $\sum \frac{1}{n^{3/2}}$  converges by  $p = 3/2 > 1$   
 $f' = \frac{1}{x^{3/2}}$

(c) Consider  $\sum_{n=1}^{\infty} h_n$  where  $h_n = 2^{f(n)/f'(n)}$ . Find  $\sum_{n=1}^{\infty} h_n$  or show that it diverges.

$h_n = 2^{\left(\frac{-2/\sqrt{n}}{1/x^{3/2}}\right)} = 2^{\left(\frac{-2x^{3/2}}{x^{1/2}}\right)} = 2^{-2x}$   $\sum 2^{-2x} = \sum \frac{1}{2^{2x}} = \sum \frac{1}{4^x}$   
 $= \sum_{x=1}^{\infty} \left(\frac{1}{4}\right)^x : \frac{1/4 \cdot 4}{1 - 1/4 \cdot 4} = \frac{1}{4-1} = \boxed{\frac{1}{3}}$

(d) Consider  $\sum_{n=1}^{\infty} k_n$  and  $\sum_{n=1}^{\infty} m_n$  where  $k_n = m_n a_n$ . If  $\sum_{n=1}^{\infty} k_n$  converges by the ratio test where

$\lim_{n \rightarrow \infty} \left| \frac{k_{n+1}}{k_n} \right| = \frac{2}{3}$ , determine whether  $\sum_{n=1}^{\infty} m_n$  converges or diverges. Explain your reasoning.

$\sum m_n = \sum \frac{k_n}{a_n} \rightarrow \lim_{n \rightarrow \infty} \left| \frac{k_{n+1}}{a_{n+1}} \cdot \frac{a_n}{k_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{k_{n+1}}{k_n} \cdot \frac{a_n}{a_{n+1}} \right|$   
 $= \frac{2}{3} \cdot \lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right| = \frac{2}{3} \lim_{n \rightarrow \infty} \left| \frac{-2/\sqrt{n}}{-2/\sqrt{n+1}} \right| = \frac{2}{3} \lim_{n \rightarrow \infty} \left| \frac{\sqrt{n+1}}{\sqrt{n}} \right| = \frac{2}{3} < 1$   
∴ converges

BC3: Let  $f(x) = -\frac{2}{\sqrt{x}}$  and let  $A = \sum_{n=1}^{\infty} a_n$  where  $a_n = f(n)$ .

$$f' = \frac{1}{x^{3/2}}$$

(e) Consider  $\sum_{n=1}^{\infty} c_n$  where  $c_n = \frac{f'(n)}{f(n)}$ . Determine whether  $\sum_{n=1}^{\infty} (-1)^{n+1} c_n$  converges or diverges.

Explain your reasoning.

$$c_n = \frac{\frac{1}{x^{3/2}}}{-\frac{2}{\sqrt{x}}} = \frac{1}{x^{3/2}} \cdot \frac{x^{1/2}}{-2} = \frac{1}{-2x}$$

Alternating Series test

$$\sum (-1)^{n+1} \cdot \frac{1}{2n}$$

i)  $\lim_{n \rightarrow \infty} \frac{1}{2n} = 0$  ? ✓

ii)  $\frac{1}{2(n+1)} < \frac{1}{2n}$  ? ✓✓

∴ **Converges**

(f) Consider  $\sum_{n=1}^{\infty} d_n$  where  $d_n = \frac{f(f'(n))}{n}$ . Determine whether  $\sum_{n=1}^{\infty} d_n$  converges or diverges.

Explain your reasoning.

$$d_n = \frac{f\left(\frac{1}{n^{3/2}}\right)}{n} = \frac{-\frac{2}{\left(\frac{1}{n^{3/2}}\right)^{1/2}}}{n} = \frac{-2n^{3/4}}{n} = -\frac{2}{n^{1/4}}$$

$\sum -\frac{2}{n^{1/4}}$  is  $p = 1/4 \leq 1$  so **Diverges** by p-series

(g) Consider  $\sum_{n=1}^{\infty} g_n$  where  $g_n = f^{-1}(n)$ . Determine whether  $\sum_{n=1}^{\infty} g_n$  converges or diverges.

Explain your reasoning.

$f(n) = -\frac{2}{\sqrt{x}}$ ,  $x = \frac{-2}{\sqrt{f(n)}} \Rightarrow \sqrt{f'(n)} = \frac{-2}{x}$

$f^{-1}(n) = \frac{4}{x^2} = g_n$

$\sum \frac{4}{x^2}$  **Converges** by  $p=2 > 1$

(h) Consider  $\sum_{n=1}^{\infty} j_n$  where  $j_n = f(n)f'(n)$ . Determine whether  $\sum_{n=1}^{\infty} j_n$  converges or diverges.

Explain your reasoning.

$$j_n = \frac{-2}{\sqrt{n}} \cdot \frac{1}{n^{3/2}} = \frac{-2}{n^2}$$

$\sum \frac{-2}{n^2}$  **Converges** by  $p=2 > 1$  p-series

**Ex. 3:** Alrighty, summary time. Determine the convergence and divergence of the following series. If the series are alternating, determine if it is conditional or absolute. If you can determine what the series converges to, find the sum:

a)  $\sum_{n=0}^{\infty} \frac{n+1}{3n+5}$

$$\lim_{n \rightarrow \infty} \frac{n+1}{3n+5} = \frac{1}{3} \neq 0$$

$\therefore$  Diverges

b)  $\sum_{n=1}^{\infty} \left(\frac{\pi}{6}\right)^{n+1}$

$|\pi/6| < 1 \dots$  Converges to 0

$$\frac{\pi^2/36}{1 - \pi/6} = \frac{\pi^2}{36 - 6\pi}$$

c)  $\sum_{n=1}^{\infty} ne^{-n^2}$  since  $ne^{-n^2}$  is positive, continuous and decreasing

$\int_1^{\infty} ne^{-n^2} dn$   $u = -n^2$   
 $du = -2n dn$

$$\lim_{b \rightarrow \infty} \left. -\frac{1}{2}e^{-n^2} \right|_1^b = \lim_{b \rightarrow \infty} \left( -\frac{1}{2}e^{-b^2} + \frac{1}{2}e^{-1} \right)$$

$$\lim_{b \rightarrow \infty} \left( \frac{1}{2e^{b^2}} + \frac{1}{2e} \right) = 0 + \frac{1}{2e} \therefore \text{converges}$$

d)  $\sum_{n=1}^{\infty} \frac{1}{\ln(n+1)}$  compare to  $\frac{1}{n+1}$

Since  $\sum \frac{1}{n+1}$  is divergent + harmonic

$$\frac{1}{\ln(n+1)} > \frac{1}{n+1} \text{ so also } \text{diverges}$$

e)  $\sum_{k=0}^{\infty} (-1)^k \frac{3}{4k+1}$

i)  $\lim_{k \rightarrow \infty} \frac{3}{4k+1} = 0? \checkmark$

ii)  $\frac{3}{4(k+1)+1} < \frac{3}{4k+1} ? \checkmark \checkmark$

$\sum \frac{3}{4k+1}$  is compared to

divergent harmonic, so

Conditional convergence

f)  $\sum_{n=1}^{\infty} \frac{n!}{10^n}$

$$\lim_{n \rightarrow \infty} \left| \frac{(n+1)!}{10^{n+1}} \cdot \frac{10^n}{n!} \right|$$

$$\lim_{n \rightarrow \infty} \left| \frac{n+1}{10} \right| = \infty > 1$$

Diverges

g)  $\sum_{n=1}^{\infty} \frac{\cos(n\pi)}{1+\sqrt{n}}$

← Alternating!

i)  $\lim_{n \rightarrow \infty} \frac{1}{1+\sqrt{n}} = 0? \checkmark$

ii)  $\frac{1}{1+\sqrt{n+1}} < \frac{1}{1+\sqrt{n}}? \checkmark \checkmark$

$\sum \frac{1}{1+\sqrt{n}}$  compares to divergent

$\sum \frac{1}{\sqrt{n}} \quad p = 1/2 \leq 1$

Conditional Convergent

i)  $\sum_{n=1}^{\infty} \frac{(-1)^{n+1} e^n}{n!}$

$\lim_{n \rightarrow \infty} \left| \frac{e^{n+1}}{(n+1)!} \cdot \frac{n!}{e^n} \right|$

$= \lim_{n \rightarrow \infty} \left| \frac{e}{n+1} \right| = 0 < 1$

$\therefore$  converges absolutely

h)  $\sum_{m=1}^{\infty} (-1)^m \frac{2m^2 - 1}{1 - m^2}$

i)  $\lim_{m \rightarrow \infty} \frac{2m^2 - 1}{1 - m^2} = 0? = -2 \times$

Diverges by  $n^{\text{th}}$  term test for divergence

j)  $\sum_{n=1}^{\infty} \frac{5^n}{3 + 6^n}$

compare to  $\sum \left(\frac{5}{6}\right)^n$

which  $|5/6| < 1$

so converges

k)  $\sum_{n=1}^{\infty} \frac{3n-1}{\sqrt[3]{n^5 + 6n + 3}}$

compare to  $\sum \frac{n}{n^{5/3}} = \sum \frac{1}{n^{2/3}}$

$\lim_{n \rightarrow \infty} \left( \frac{3n-1}{\sqrt[3]{n^5 + 6n + 3}} \cdot \frac{1}{n^{2/3}} \right) = \lim_{n \rightarrow \infty} \left( \frac{3n^{5/3} - n^{7/3}}{\sqrt[3]{n^5 + 6n + 3}} \right)$

$= 3 \therefore$  since  $p = 2/3 \leq 1$

Diverges, this also Diverges

l)  $\sum_{n=1}^{\infty} \left( \frac{2}{n+1} - \frac{2}{n+3} \right) = \sum \frac{4}{(n+1)(n+3)}$

compare to  $\sum \frac{1}{n^2}$

which  $p = 2 > 1$

Converges